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PART-B

6. HDFS :- HDFS refers to Hadoop Distributed File System which is used to store data of large amount across multiple clusters.

→ In Traditional file systems, the storage of large datasets will be difficult and requires more memory space in the system

→ To overcome this problems, Hadoop System is introduced, which consists HDFS to store large datasets across multiple clusters on commodity hardware.

HDFS daemons :- Daemons is the processes running in the background.

→ Daemons means process

→ HDFS daemons are set of processes run in the Hadoop System.

→ The Hadoop 2 consists of following daemons:

i. Name Node → 50080 port

ii. Data Node → 50082 port

iii. Secondary Name Node → 50090 port

iv. Resource Manager → 8088 port

v. Node Manager → 8042 port

→ The HDFS consists NamNode (Master System) and DataNode (Slave System)

→ The NameNode, Secondary Name Node, Resource Manager works on Master System and

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DataNode, NodeManager works on slave system.

i. Name Node:

→ NameNode works on Master System.

→ The Primary purpose of NameNode is to store the Metadata.

→ Metadata is the list of files in HDFS.

→ The NameNode will interact with datanode, at which locations of block of files store in Metadata.

→ The Metadata is stored in memory.

Starting of NameNode:-

hadoop -daemon.sh start namenode

stoping of NameNode:-

hadoop -daemon.sh stop namenode

ii. Data Node:

→ Data Node works on slave system.

→ The NameNode always interact with data node.

→ The Data Node is actual worker, follows the instructions of Namenode.

→ Data Node will read, write and process the data in files.

starting of DataNode:-

hadoop -daemon.sh start datanode

stoping of DataNode:-

hadoop -daemon.sh stop datanode

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iii. Secondary Name Node (SNN):

→ The Secondary Name Node used for hourly backup.

→ In Hadoop if files are failed or crashes then the secondary namenode is used for hourly backup to recover the data.

→ The recovered data is stored in a file named fsimage and create new system, new datanode to store blocks.

→ Functionalities SNN :-

> It groups the edit logs and fsimage together.

> It ensures the transactional logs.

→ It uses checkpoints to check the data so SNN known as checkpoint Node.

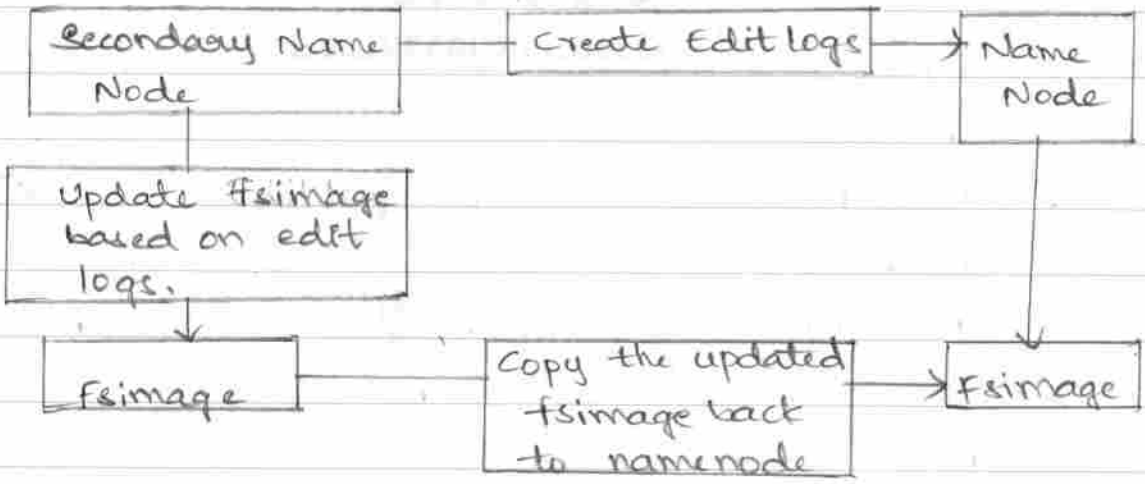

fig: Secondary Name Node.

iv. Resource Manager:

→ Resource Manager works on Master System.

→ Manages the resources on various applications.

→ It consists of two components:

i. Application Manager: responsible for accepting

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request for client.

ii. Scheduler: utilized to provide resources across applications.

Starting of Resource Manager:-

yarn-daemon.sh start resourcemanager

stoping:-

yarn -daemon.sh stop resourcemanager

v. Node Manager:

→ Works on slave system. Manages memory resources in node and memory.

Starting:- yarn -daemon.sh start nodemanager

Stoping:- yarn -daemon.sh stop nodemanager

→

8. Hadoop Distributed File System (HDFS):-

Hadoop is an open source framework for store and process the large volume of data.

→ As the usage increasing, the limit of data increases which made difficult to the traditional file systems.

→ In this situation Hadoop came into picture, which has HDFS, YARN, MapReduce and HBase as core components for storing, processing and analyzing large datasets across multiple clusters on commodity hardware.

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→ HDFS is the distributed file system which stores files in distributed manner consumes less space.

→ HDFS is the backbone of the Hadoop System.

→ HDFS is the system where files are stored in clear manner either structured or unstructured data.

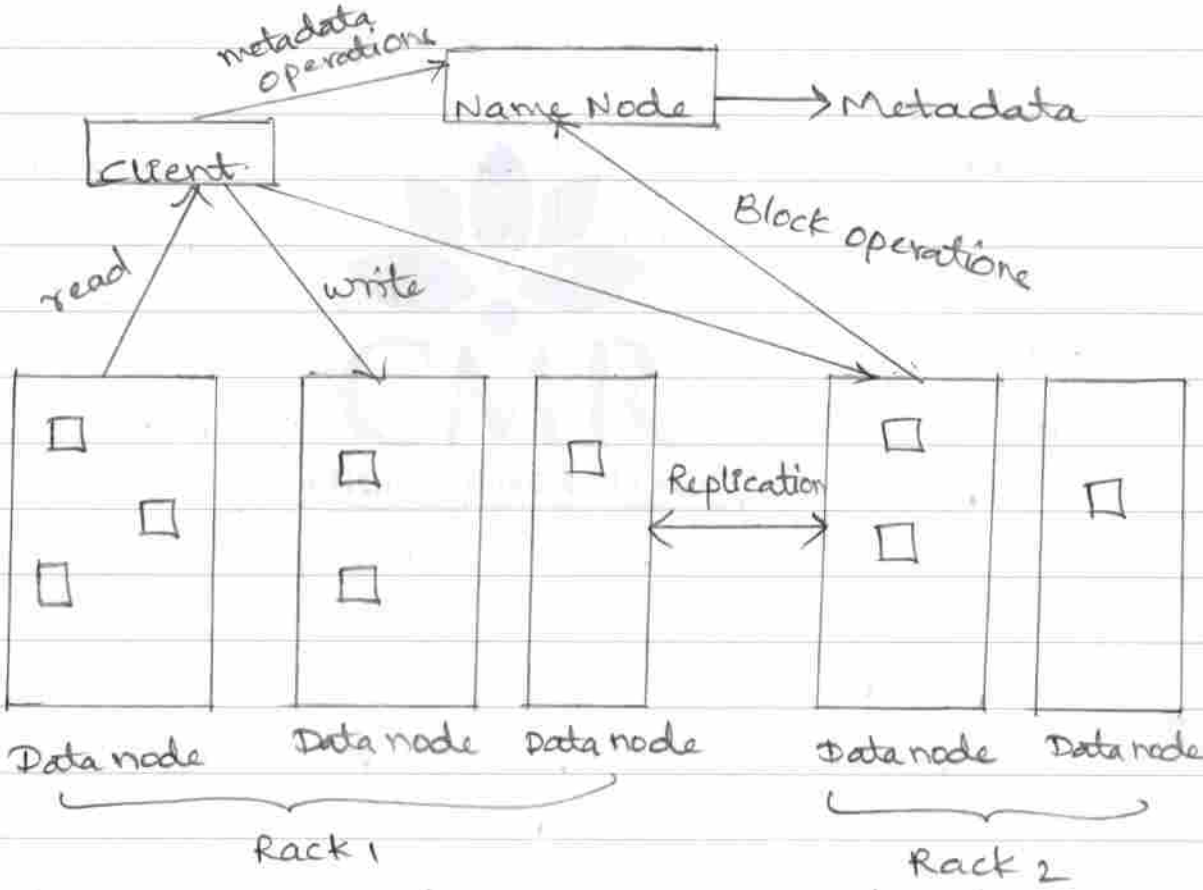

fig.: HDFS Architecture

→ The HDFS is made up of NameNode, DataNode, Secondary Name Node, checkpoints, Backup node and Blocks.

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Components of HDFS :-

i. NameNode :

→ NameNode is the Master Node.

→ NameNode manages all datanodes and assign works to them.

→ The Namenode won't read data that exists in file.

→ Namenode executes filesystem namespaces, opening, closing and renaming files and directories.

→ Namenode is to deploy on reliable hardware which has high configuration, not on commodity hardware.

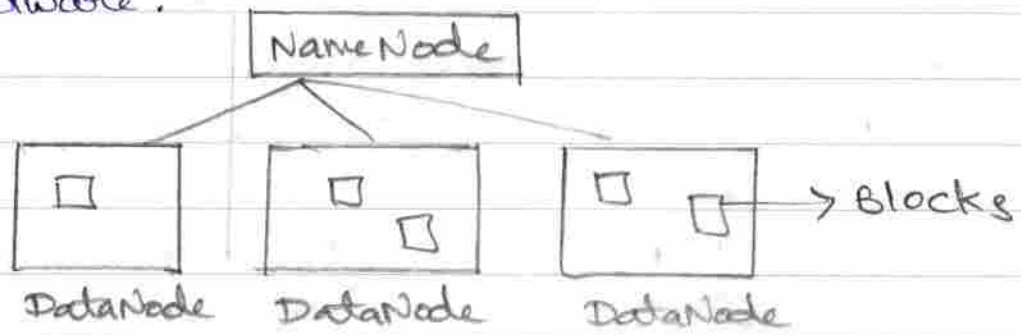
ii. DataNode :

→ DataNode is the actual worker.

→ DataNode will follow the instructions of NameNode and perform operations such as creation, deletion etc.

→ DataNode will perform reading, writing & processing, etc.

→ DataNode is deployed on commodity hardware.



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iii. Secondary Name Node:

→ The Secondary Name Node (SNN) is to merge edit logs and fsimage together.

→ The Secondary Name Node will generate checkpoints so it is also known checkpoint node

→ In this node all transaction logs are stored and transfers as needed.

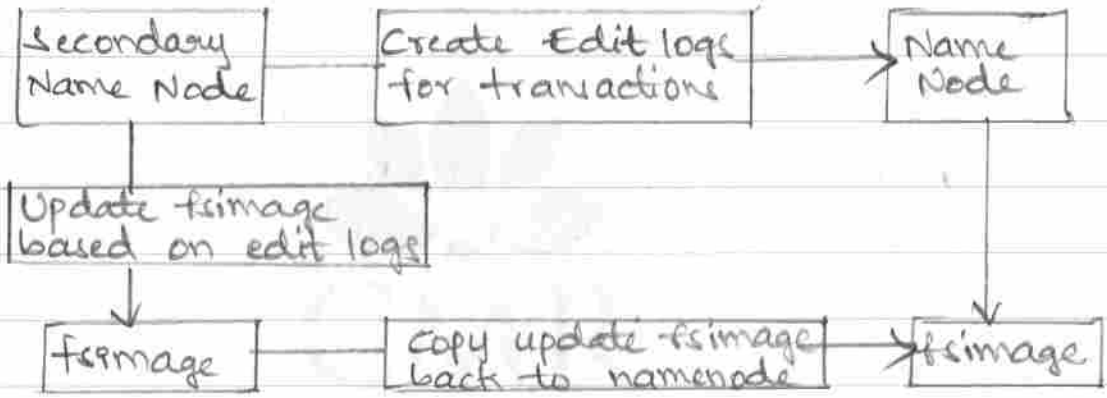

fig.: Secondary Name Node

iv. Checkpoints:

→ Checkpoints are generated when the edit logs and fsimage merges.

→ For every transaction a new checkpoint is generates by SNN.

v. Backup Node:

→ Backup Node is to active the NameNode or DataNode in case of NameNode fails.

→ But, can not use/replace NameNode with Backup Node.

vi. Blocks: Blocks are the primary locations were data stores in the HDFS.

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10. Machine learning :-

Machine learning is a subfield of Artificial Intelligence which allows models to learn from patterns and predefined classes to predict the outcomes.

→ In simple, Machine learning is a process of making model to learn from data and make decisions like humans to process and analyze the data.

Various techniques of Machine learning :-

The machine learning has categorized into follow :

i. Supervised learning
ii. Unsupervised learning
iii. Reinforcement learning
iv. Semi-Supervised learning

i. Supervised learning :

→ Supervised learning is defined as models trained by labelled.

→ Labelleds are combination of input features and output labels.

→ In this learning, learns to map points between input and correct output.

→ It is both validation and training labelled.

→ There are two types :

a. Classification b. Regression.

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a) Classification : is predicts the outcomes based on predefined classes.

> Decision Tree, Logistic Regression are algorithms of classification.

b) Regression : is reading continuous data, it finds the relationship between ~~clusters~~ inputs and numerical value.

> Linear regression, polynomial regression etc are algorithms.

Eg.: When you train a model with labelled images of cats & dogs, it learns the features from them.

When you shown new image, it predicts the outcome correctly.

II. Unsupervised learning:

→ Unsupervised learning is the process of training model with unlabelled data.

→ Unlabelled data means without output labels.

→ There are few types :

a) Clustering : is a grouping of similar items based on similarities into clusters

b) Dimensionality Reduction Technique : is to reduce no. of features while storing important information.

c) Association rules : to determine the behavior or assign.

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Eg.: When you have customers unlabelled data, you can seggregate them into clusters based on their purchase behavior.

iii. Reinforcement learning:

→ Reinforcement learning is the training agent to produce sequence of transaction through trail and errors.

→ This technique will receive feedback in the form of reward / penalities to over the optimal time.

→ There are two types:

a) Positive Reinforcement

b) Negative Reinforcement

→ Algorithms used such as

> Q-learning

> Deep Q-learning

Eg.: When a Agent learns to play chess, it gets positive feedback for positive move and negative feedback for bad move, anyway learns strategy to win.

iv. Semi-supervised learning:

→ It is the combination of both supervised and unsupervised learning.

→ It takes small amount of labelled datasets and large amount of unlabelled datasets, to get efficient in costly systems.

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Eg.: When you train translation model, language translation takes labelled sentence pair gives intensive outcomes. In this both labelled & unlabelled sentence pair is provided to test the model.

11. Social Media Analytics :-

Social Media Analytice are the extracting, deriving insightful information from the large volume of social media data.

→ Social Media generates enormous of data in the form of likes, shares, posts and messages.

→ The population increasing, so as the social media usage also increasing thoroughly.

→ The social media analytics will help to brand promotions, influences, audience interactione.

→ The social media will be positive as well as negative effect to the society.

→ Social media can be twitter, facebook, instagram, linkedin, whatsapp etc.

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Importance of Social Media Analytics :-

→ Social Media Analytics are important to know the trends going on world.

→ This will help to promote their brands as positive or negative talks will spread wildefire through social media.

→ And also it will helps to know the strategies for business.

→ shows the competator tricks and way of improving business.

→ Social Media Analytics also allows to read the data by "retweet" or "tweetr".

→ Similarly can also used for other social media platforms.

→ By Social Media Analytics every small issue can be spread worldwide and solves within in days.

→ If we take an example of facebook, there many videos will be uploaded as seeking for help or raising their issues at their places.

→ These all problems will reach to higher authorities or helpers through shares, likes and reposts then the

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issues get resolved.

Applications:-

1. Content Sharing: The Social Media Analytics are used to share the content to one another through connections, friends etc.

2. Audience Support: The audience are support and increase the usage of social media.

3. Paid Influence: Influences the content based on investment.

4. Fair Influence: Checks the content genuinly influenced, reached high level through good content & quality.

→ In hadoop Social Media Analytics will follow HDFS to store the data across multiple places.

→ Which also update and adjust the data real-time as data produces regularly.

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PART-A

1. Inputs and Outputs of MapReduce:

→ Inputs are which streamed into key-value pairs by Map() in MapReduce.
→ Outputs are which takes result of map and streams the data in key value pairs by Reduce().of MapReduce.

2. Data moved in and out of Hadoop:-

→ Data moved in of Hadoop is, ingestion of data into HDFS from various sources.
→ Data moved out of Hadoop is, exgress of data into various sources such as warehouse, from HDFS

3. Function of Secondary Name Node:-

Secondary Name Node will merge the edit logs and fsimage together, make scalable.
→ secondary NameNode will generate checkpoints when edit logs and fsimage merges newly.

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4. Supervised Learning :-

Supervised learning is the trained a model with labelled data (input and corresponding outputs) to get correct outcomes.

Eg.: When you train model with labelled images cat & dogs it learns features and if you shown new, it will identify.

5. Applications of R :-

→ Data analysis in R

→ Data Serialization & visualization

Eg.: Box plate, Sand box, gplot